Alternate Circuit Representation

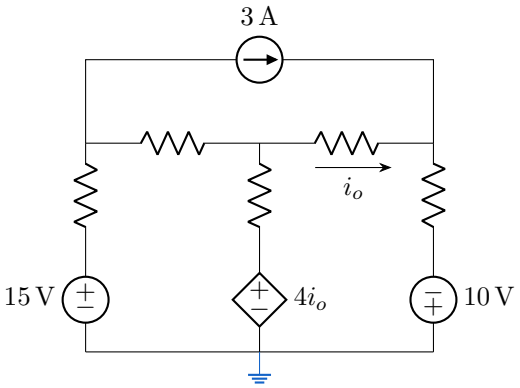
Line representation, nodal recap, voltage divider and KVL in a line

The alternate circuit diagram, or *line representation*, helps us draw circuits in a compact form. Plus, in real-life designs, this type of representation is preferred.

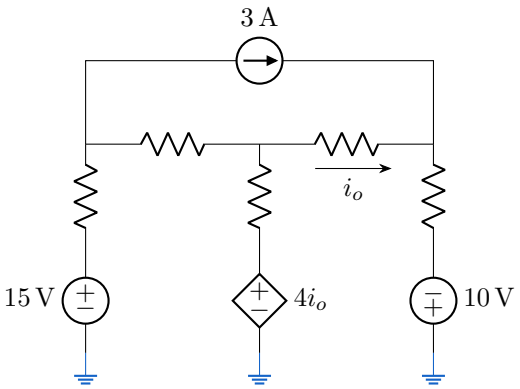
To change a regular circuit representation to line representation, follow the steps below.

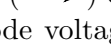
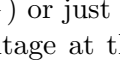
Steps

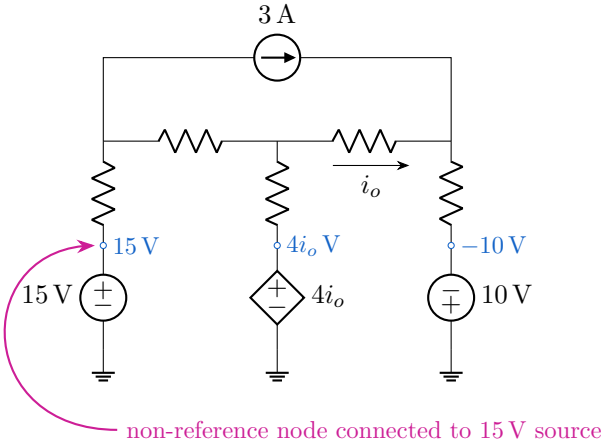
Step 0. Place a ground in the circuit. Ignore this step if a ground is already present.
To select the ground node, just follow the rules of thumb introduced in nodal analysis during your 250 course.



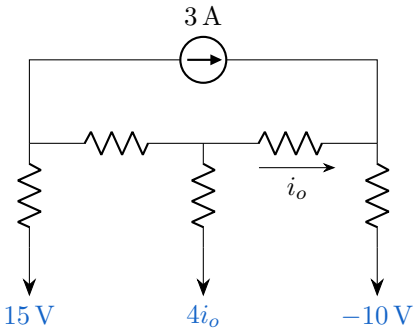
Step 1. Separate the grounded points.
Note that these points are still the same node.



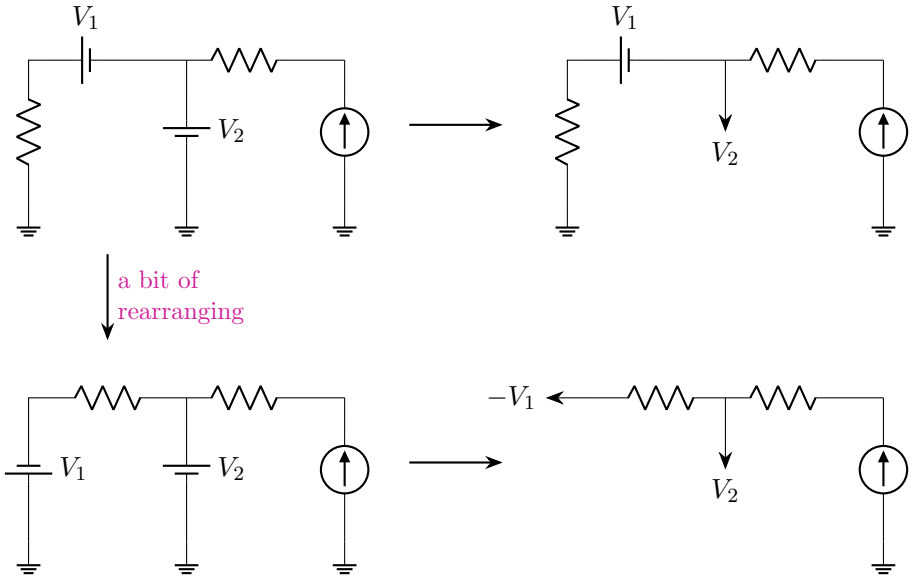
Step 2. Replace all the voltage sources connected to the ground ( ) with an arrow () or just a dot ().
Assign the node voltage at the non-reference node connected with these sources to the arrow or the dot.



Note. You cannot convert current sources; neither can you convert non-grounded voltage sources. So keep them as they are.



One exception

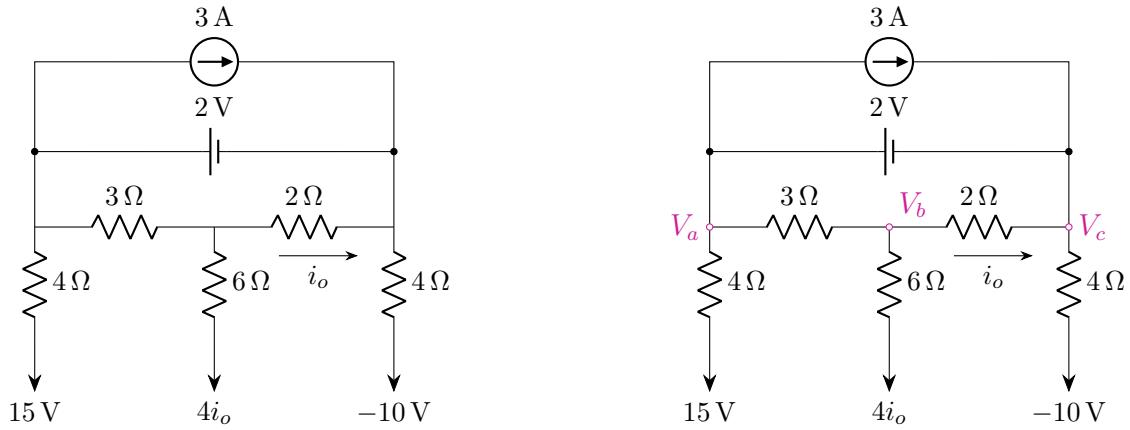


Sometimes a bit of circuit rearrangement can allow us to further simplify the circuit in alternate representation.

Should you do it? ⟶ It's up to you. It's fine even if you don't do it.

Nodal recap

Find all node voltages in the circuit.



1. Place voltage variables at all essential nodes with unknown node voltages (right figure).

2. Write KCL equations for all the marked nodes and supernodes. Supernodes will also give you

voltage relationship equations.

Voltage relationship equation @ SN a, c:

$$V_a - V_c = 2 \tag{i}$$

KCL @ SN a, c:

$$\frac{V_a - 15}{4} + \frac{V_a - V_b}{3} + \frac{V_c - V_b}{2} + \frac{V_c - (-10)}{4} = 0 \qquad \text{[3 A source shorted between the SN]}$$

$$\Rightarrow\ V_a\left(4^{-1} + 3^{-1}\right) - V_b\left(3^{-1} + 2^{-1}\right) + V_c\left(2^{-1} + 4^{-1}\right) = \frac{15}{4} - \frac{10}{4} \tag{ii}$$

KCL @ b:

$$\frac{V_b - V_a}{3} + \frac{V_b - 4i_o}{6} + \frac{V_b - V_c}{2} = 0$$

In the circuit, $i_o = \dfrac{V_b - V_c}{2}$. Therefore,

$$\frac{V_b - V_a}{3} + \frac{V_b - 4i_o}{6} + \frac{V_b - V_c}{2} = 0 \ \Rightarrow\ \frac{V_b - V_a}{3} + \frac{V_b - 2V_b + 2V_c}{6} + \frac{V_b - V_c}{2} = 0$$

$$\Rightarrow\ -V_a\left(3^{-1}\right) + V_b\left(3^{-1} - 6^{-1} + 2^{-1}\right) - V_c\left(-3^{-1} + 2^{-1}\right) = 0 \tag{iii}$$

3. Solve the equations to get the values of V_a, V_b and V_c.

Some key points

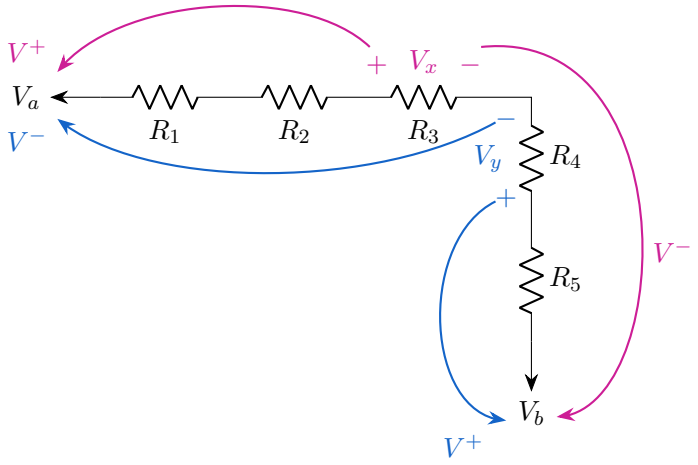
(i) If current enters the node $\longrightarrow$ negative term;
if current leaves the node $\longrightarrow$ positive term.

(ii) For resistive branches, always consider outward current.

(iii) Always replace extra variables in terms of node variables.

Voltage divider

For any series string of resistances:

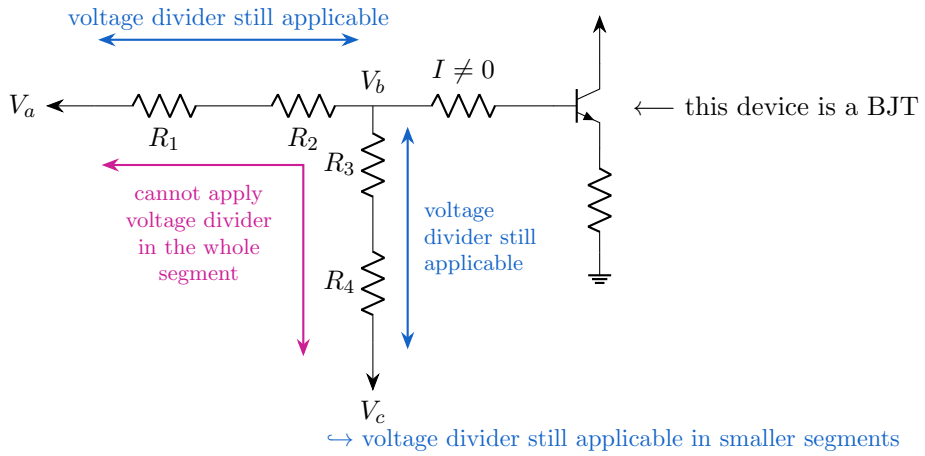


follow the colours.

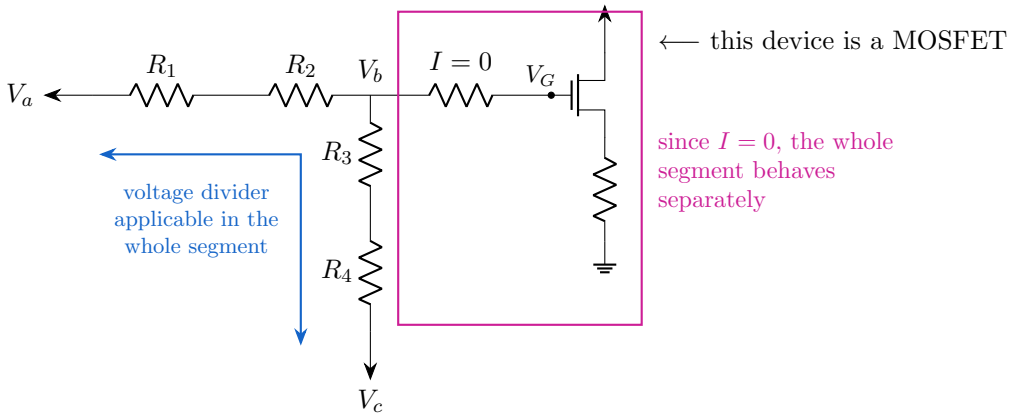
$$V_x = \frac{R_3}{\sum R}\left(V^+ - V^-\right) = \frac{R_3}{\sum R}\left(V_a - V_b\right)$$

$$V_y = \frac{R_4}{\sum R}\left(V^+ - V^-\right) = \frac{R_4}{\sum R}\left(V_b - V_a\right)$$

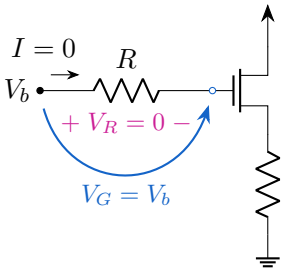
In case there is a branch connected to the string of resistors like this:



However, for certain devices:



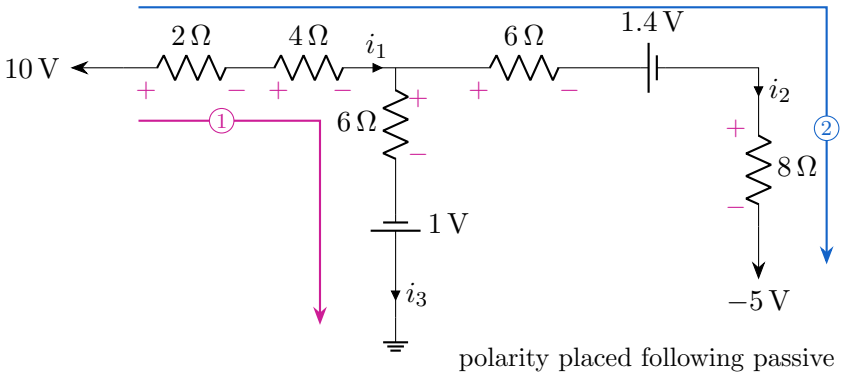
Since $I = 0$ in the connecting branch, the node voltage V_b can be transferred inward:



This concept will be tested multiple times in the course.

KVL in a line

While the voltage divider will help you out in some of the tricky cases, KVL will remain a staple in the harder (read: lengthy) but straightforward problems.
Unlike 250, where we applied KVL in a loop, we will apply KVL in lines for 251.



polarity placed following passive sign convention.

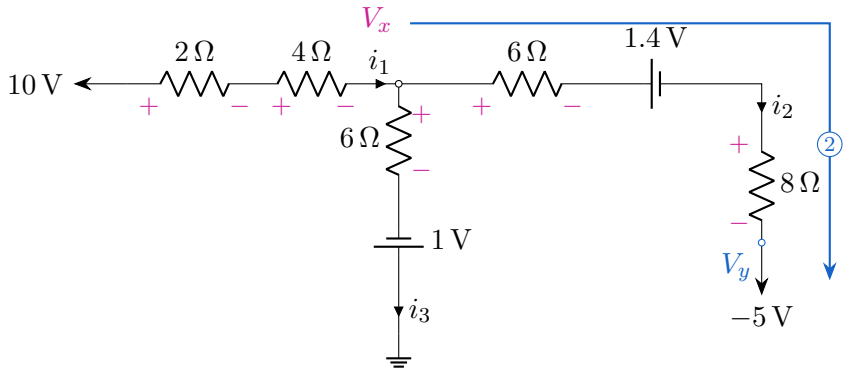
KVL in line ①:

$$2i_1 + 4i_1 + 6i_3 - 1 = V_{\text{start}} - V_{\text{end}}$$
$$\Rightarrow\ 6i_1 + 6i_3 = 10 - 0 + 1\ \Rightarrow\ 6i_1 + 6i_3 = 11$$

KVL in line ②:

$$2i_1 + 4i_1 + 6i_2 + 1.4 + 8i_2 = V_{\text{start}} - V_{\text{end}}$$
$$\Rightarrow\ 6i_1 + 14i_2 = 10 - (-5) - 1.4\ \Rightarrow\ 6i_1 + 14i_2 = 13.6$$

KVL in a line is more flexible than regular KVL. Let's take the above example and update line 2. We can update the span of KVL freely.



KVL in line 2 starting from node x to y

optionally you can specify the range → I generally do it.

Going ahead, most circuits will have named nodes, so you will not have to name them all.

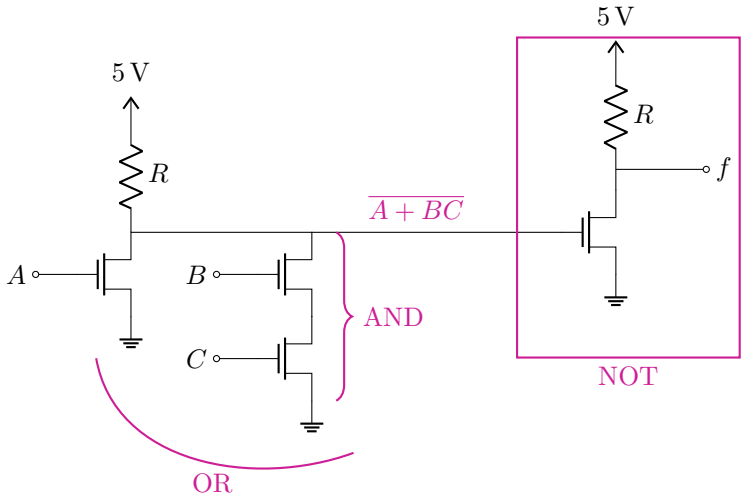
$$6i_2 + 1.4 + 8i_2 = V_{\text{start}} - V_{\text{end}}$$
$$\Rightarrow\ 14i_2 = V_x + 5 - 1.4\ \Rightarrow\ 14i_2 = V_x + 3.6$$

Some extra notes

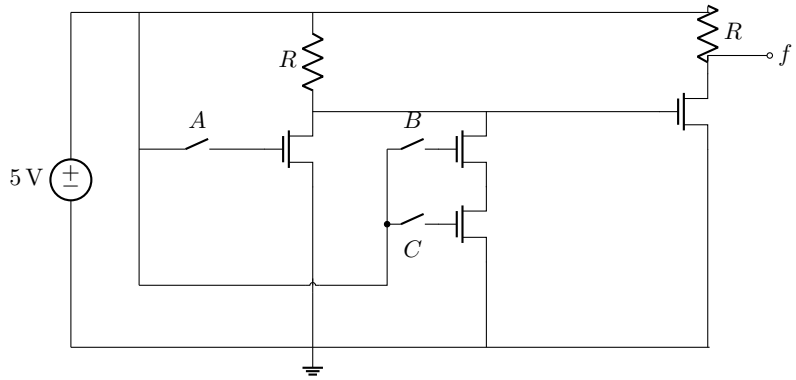
Why do we need alternate representation?

Let me present a small logic function implemented with a circuit:

$$f = A + BC \qquad [\,A \text{ OR } (B \text{ AND } C)\,]$$



If you want to convert it to 250's representation, it will look something like this:

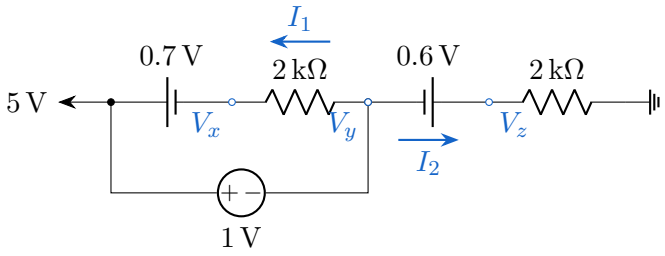


This is a very optimized representation, and even then it feels a bit cluttered and is harder to explain. You can try lengthier circuits (later, when we study them) and you will appreciate line representation more.

One last note. I personally treat all 251 circuits like puzzles and generally try to solve them by node hopping, a bit of Ohm's law and a bit of KVL, but I do realize most people find nodal analysis to be easier. I would suggest you find your own comfort zone. I have left all the assignment problems open-ended except Q1, so feel free to go ham. If you are struggling with any of the maths, you can always consult me.

Worked examples

For the given circuit, find V_x, V_y, V_z, I_1 and I_2.

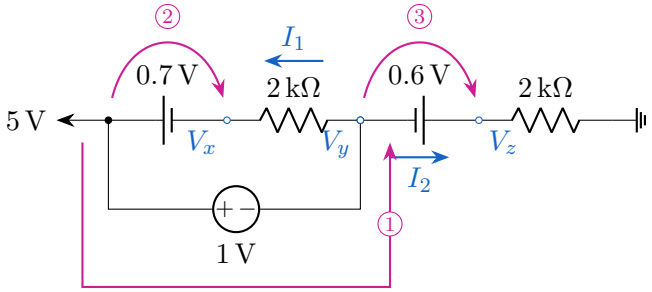


Solution.

① → $V_y = 5 - 1 = 4\,\text{V}$
② → $V_x = 5 - 0.7 = 4.3\,\text{V}$
③ → $V_z = V_y + 0.6 = 4.6\,\text{V}$

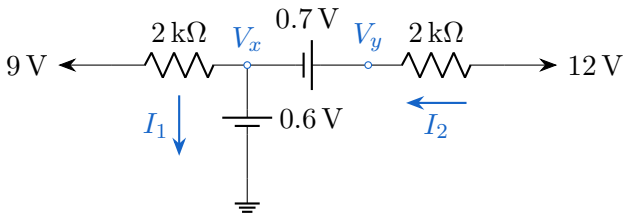
$$I_1 = \frac{V_y - V_x}{2} = \frac{4 - 4.3}{2} = -0.15\,\text{mA}$$

$$I_2 = \frac{V_z - 0}{2} = \frac{4.6}{2} = 2.3\,\text{mA}$$



The arrows represent node voltage transfer paths.
You do NOT have to show the paths. I am showing them for you to understand.

For the following circuit, find V_x, V_y, I_1 and I_2.



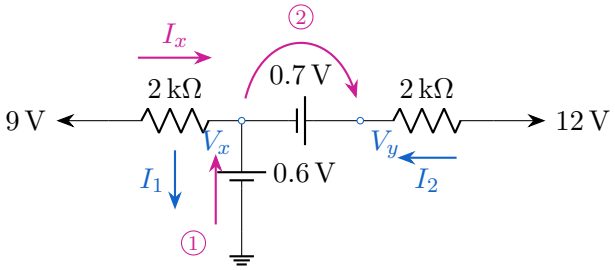
Solution.

① $V_x = 0.6\,\text{V}$
② $V_y = V_x + 0.7 = 1.3\,\text{V}$

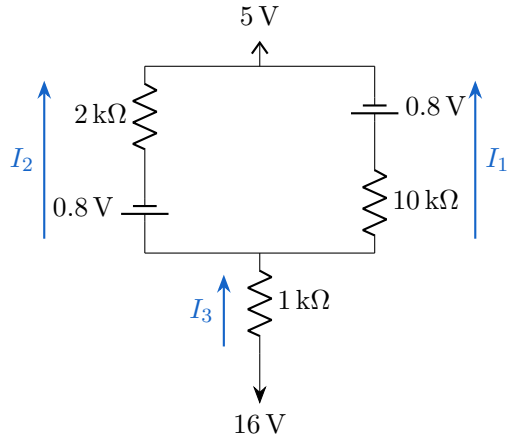
$$I_x = \frac{9 - V_x}{2} = \frac{8.4}{2} = 4.2\,\text{mA}$$

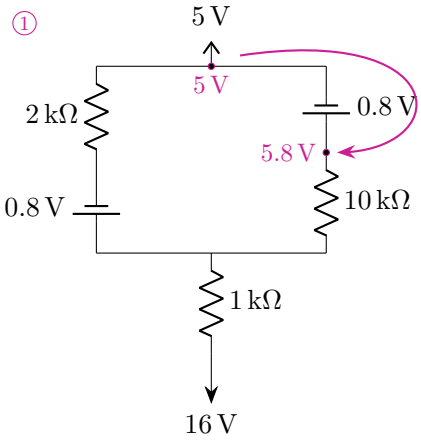
$$I_2 = \frac{12 - V_y}{2} = \frac{10.7}{2} = 5.35\,\text{mA}$$

$\therefore\ I_1 = I_x + I_2 = 9.55\,\text{mA}$

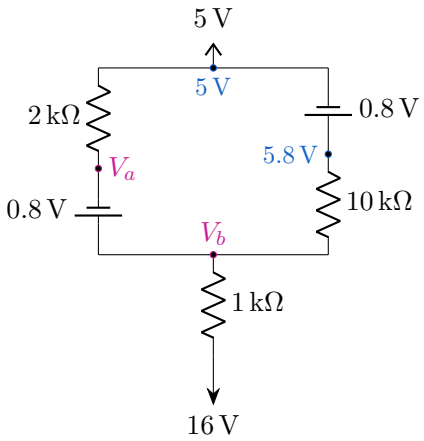


Find I_1, I_2, I_3 and all the node voltages.





Using nodal analysis



KCL @ SN a, b:

$$\frac{V_a - 5}{2} + \frac{V_b - 5.8}{10} + \frac{V_b - 16}{1} = 0$$

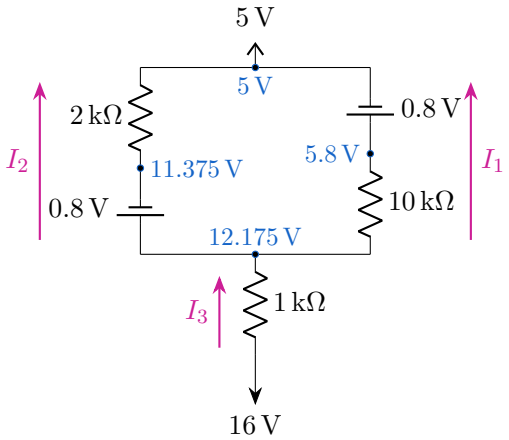
$$\Rightarrow \; V_a\left(2^{-1}\right) + V_b\left(10^{-1} + 1^{-1}\right) = 2.5 + 0.58 + 16$$

Voltage relation equation @ SN a, b:

$$-V_a + V_b = 0.8$$

Solving the equations:

$$V_a = 11.375\,\text{V}, \qquad V_b = 12.175\,\text{V}$$

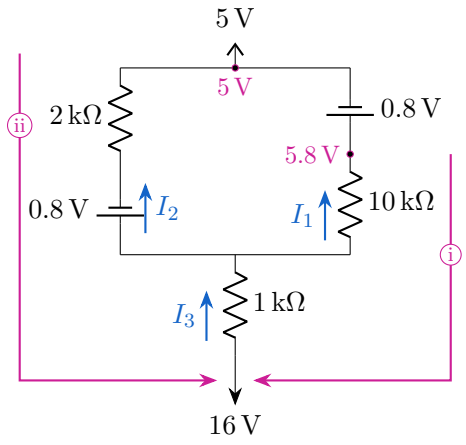


$$I_2 = \frac{11.375 - 5}{2} = 3.1875\,\text{mA}$$

$$I_1 = \frac{12.175 - 5.8}{10} = 0.6375\,\text{mA}$$

$$I_3 = \frac{16 - 12.175}{1} = 3.825\,\text{mA}$$

Transferring node voltages



KVL in line ⓘ:

$$-10I_1 - I_3 = 5.8 - 16$$

KVL in line ⓘⓘ:

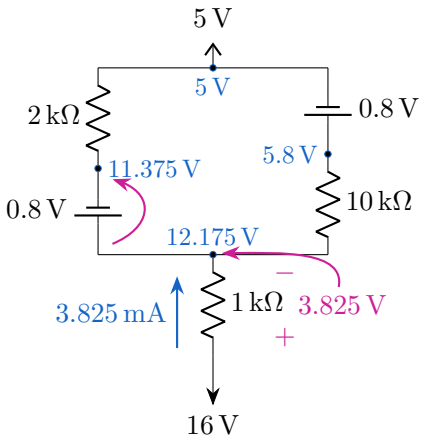
$$-2I_2 - 0.8 - I_3 = 5 - 16$$

Using KCL:

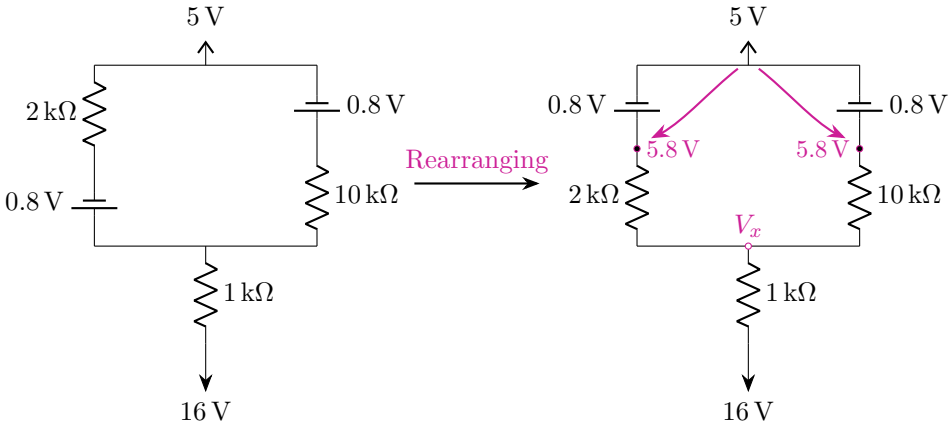
$$I_3 = I_1 + I_2$$

Solving the equations we get:

$I_1 = 0.6375\,\text{mA}, \quad I_2 = 3.1875\,\text{mA}, \quad I_3 = 3.825\,\text{mA}$



You can make the nodal analysis even shorter



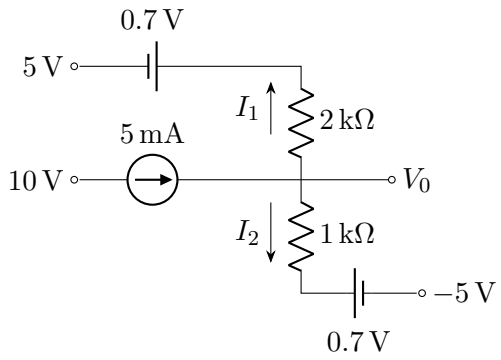
KCL @ node x:

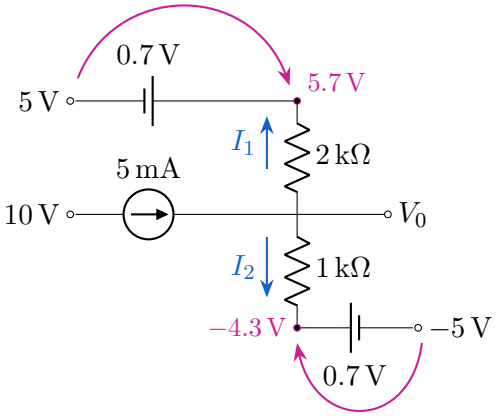
$$\frac{V_x - 5.8}{2} + \frac{V_x - 5.8}{10} + \frac{V_x - 16}{1} = 0 \qquad \therefore\ V_x = 12.175$$

Then just find I_1, I_2 and I_3 as usual.

Takeaway: There are multiple ways to solve the same problem.

Find I_1, I_2 and V_0.





Using KCL @ the output node:

$$5 = I_1 + I_2 \;\rightarrow\; 5 = \frac{V_0 - 5.7}{2} + \frac{V_0 - (-4.3)}{1}$$

$$\therefore\; V_0 = 2.367\,\text{V}$$

$$\therefore\; I_1 = \frac{2.367 - 5.7}{2} = -1.6665\,\text{mA}$$

$$I_2 = \frac{2.367 - (-4.3)}{1} = 6.667\,\text{mA}$$